

Program Summary - THE1.sas

Execution Environment

Author: u64127570
File: /home/u64127570/sasuser.v94/THE1.sas
SAS Platform: Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host: ODAWS01-USW2-2.ODA.SAS.COM
SAS Version: 9.04.01M7P08062020
SAS Locale: en_US
Submission Time: 4/17/2025, 10:38:12 PM
Browser Host: C-67-165-187-91.HSD1.IL.COMCAST.NET
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0
Application Server: ODAMID00-USW2-2.ODA.SAS.COM

Code: THE1.sas

```
/* Take Home Exam - Q1 */
```

Log: THE1.sas

Notes (4)

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;  
68  
69      /* Take Home Exam - Q1 */  
70  
71      data minor;  
72      infile "/home/u64127570/sasuser.v94/DelinquencyandIQ.txt" dlm='09'x dsd missover firstobs=2;  
73      input IQ DI;  
74      run;
```

NOTE: The infile "/home/u64127570/sasuser.v94/DelinquencyandIQ.txt" is:
Filename=/home/u64127570/sasuser.v94/DelinquencyandIQ.txt,
Owner Name=u64127570,Group Name=oda,
Access Permission=-rw-r--r--,
Last Modified=17Apr2025:10:11:56,
File Size (bytes)=156

NOTE: 16 records were read from the infile "/home/u64127570/sasuser.v94/DelinquencyandIQ.txt".
The minimum record length was 5.
The maximum record length was 9.

NOTE: The data set WORK.MINOR has 16 observations and 2 variables.

NOTE: DATA statement used (Total process time):
real time 0.03 seconds
user cpu time 0.00 seconds
system cpu time 0.00 seconds

```

memory          756.43k
OS Memory       23460.00k
Timestamp       04/18/2025 03:38:11 AM
Step Count      38   Switch Count  2
Page Faults     0
Page Reclaims   121
Page Swaps      0
Voluntary Context Switches 18
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 272

```

```

75
76      /* proc print data = minor; */
77      /* run; */
78
79      proc reg data = minor;
80      title "Linear Regression Predicting DI from IQ";
81      model DI = IQ /clb;
82      output out=minor_reg p= pred_DI;
83      run;
84
85      /* DI = -0.43801(IQ)+72.89252
86      Rsq= 0.7032 Rsqadj= 0.6821
87      70.32% of the variation in DI is explained by IQ in this model
88      Slope parameter is not 0: t = -5.76 p <.0001
89      null: IQ does not effect DI
90      alt: IQ effects DI
91      At a 0.05 significant level, we reject the null hypothesis, IQ significantly effects DI
92      CI: -0.60112, -0.27490. We are 95% confident that the true slope parameter lies between -0.60112, -0.27490.
93      The negative CI suggests a negative relationship between IQ and DI, as IQ increases, DI decreases.
94      The residuals plots look good enough. residual V predicted value looks pretty random but there is an argument for slight
95      ! funneling (cant confirm without more data, can just be an outlier), residual v percent is normal enough.
96      RStudent v predicted value has one point (>3) above the threshold of 2, also one point above the cooks d threshold
97      ! (0.25), but its leverage is quite low.
98      So, one outlier, but its leverage is low enough that it likely is not influencing the coefficents very much.
99      */
100
101      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
102
103
104
105
106
107
108

```

Results: THE1.sas

Linear Regression Predicting DI from IQ

The REG Procedure
Model: MODEL1
Dependent Variable: DI

Number of Observations Read	16
Number of Observations Used	16

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	653.64229	653.64229	33.17	<.0001
Error	14	275.87271	19.70519		
Corrected Total	15	929.51500			

Root MSE	4.43905	R-Square	0.7032
Dependent Mean	28.32500	Adj R-Sq	0.6820
Coeff Var	15.67186		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	72.89252	7.81735	9.32	<.0001	56.12597 89.65906
IQ	1	-0.43801	0.07605	-5.76	<.0001	-0.60112 -0.27490

Linear Regression Predicting DI from IQ

The REG Procedure
Model: MODEL1
Dependent Variable: DI



